

GEMINI TOKYO HACKATHON 2026

SOCCKER STRIKER

AI agents play the match. You step in at the moments that matter.

A real-time, agent-driven football game

WHY THIS APP

Make AI agents that actually play football

- ◆ **The moment:** the 2026 World Cup in the USA — football has the world's attention.
- ◆ **The idea:** agents that genuinely play and cooperate, not scripted bots.
- ◆ **For everyone:** natural motion control, like Nintendo Wii Sports — swing the iPhone to play.

WHAT IT IS

4 vs 4. Both teams are autonomous AI agents.

- ◆ Mac is the pitch — the full 3D match runs here.
- ◆ iPhone is the controller — a motion controller in your hand.
- ◆ You don't micromanage. You intervene at decisive moments by swinging your iPhone.

THE HARD PROBLEM

An LLM cannot drive a 60fps game

Calling a model every frame is impossible — network latency runs from hundreds of milliseconds to seconds, and the cost is prohibitive.

So how can a Gemini agent genuinely play in real time?

STRENGTH 1 — GEMINI AS THE BRAIN

Hybrid agent loop

Gemini 3.5 Flash decides intentions every ~1.5s — per player: move / mark / pass / shoot / support

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60Hz local engine executes those intentions every frame

Gemini is the decision-maker; the engine is the muscle. The game never stalls waiting on the model, and it stays in character.

STRENGTH 2 — BELIEVABLE TEAM PLAY

Real-time multi-agent behavior

- ◆ Possession model, player roles, **steering with separation** (no clustering), man-marking, pass-lane checks, off-ball runs.
- ◆ Pure-Swift, **deterministic engine** — covered by 16 unit tests.
- ◆ Runs fully offline and **gracefully falls back to rule AI** if the API is unavailable.

STRENGTH 3 — HUMAN IN THE LOOP

Motion control, Wii-Sports style

- ◆ **iPhone motion controller:** CoreMotion at 100Hz — swing to kick.
- ◆ Connects to the Mac over **Bonjour + UDP peer-to-peer** — zero setup.
- ◆ Decisive moments trigger **timing and power gauges**; nail it and the manager reacts: "tactics on point."

WHERE WE PUT THE EFFORT

Craft that sells the fantasy

- ◆ 3D pitch and players in **Three.js** (WebView) — run animation, uniforms per nation.
- ◆ Nation select with real flags and player models; **cut-in animations** for shots, dribbles, and the manager.
- ◆ Audio: procedural SFX plus stadium ambience. Gemini key handled via an **.env-style xcconfig**.

STACK

How it is built

◆ AI

Gemini 3.5 Flash — Gemini API,
agentic, JSON-structured
output

◆ App

Swift 6 / SwiftUI — macOS game
+ iOS controller, shared package

◆ Render

Three.js / WebKit

◆ Motion & Net

CoreMotion,
Network.framework
(Bonjour/UDP)

VISION

Agents play. Humans join in.

Next: 11 vs 11, deeper tactics direct from the agent, more nations, and online play.